Project Report

AI-Driven Threat Modeling for Resilient Cyber-Physical Systems: Learning-Aware Adversarial Risk Analysis and Mitigation

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Cyber Security using Threat Modeling and Attack Simulation (CTMAS)

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Abstract

Cyber–physical systems (CPS) that operate critical infrastructure face tightly coupled cyber and process risks, where a security compromise can propagate directly to unsafe physical behavior. Static design-time threat modeling offers structured visibility into attack surfaces, but it lacks runtime awareness. Purely data-driven anomaly detection offers timely alerts, but often provides limited threat semantics and weak operational interpretability. This project addresses that gap through an integrated framework for *AI-driven threat modeling* over the Secure Water Treatment (SWaT) testbed. The proposed workflow begins with CPS architecture analysis, asset identification, STRIDE-based threat enumeration, and attack-tree construction. It then preprocesses SWaT telemetry into learning-ready windows, trains a stacked Long Short-Term Memory (LSTM) model for attack behavior learning, and fuses the learned attack probability signal with STRIDE baseline likelihoods to obtain time-varying threat risk. A comparative analysis is performed across standalone LSTM reasoning, LSTM+PASTA, and LSTM+STRIDE configurations, while MITRE ATT&CK for ICS mapping and a lightweight threat ontology are used to produce contextual narratives and operator-friendly outputs.

The implemented pipeline processes 946,728 cleaned SWaT samples, uses 51 telemetry features, and learns from 30-step temporal windows. The trained LSTM achieves an accuracy of 99.57%, precision of 93.34%, recall of 99.65%, F1-score of 96.39%, ROC–AUC of 0.9999, and PR–AUC of 0.9985. The integrated risk model raises system risk from a static baseline of 6.574 to an average of 7.830 during attack periods, indicating strong adaptive prioritization under adversarial conditions. Comparative evaluation further shows that the hybrid framework preserves the strong detection capability of the neural detector while adding dynamic threat ranking, richer explainability, and higher resilience scoring. Overall, the project demonstrates a complete and technically coherent workflow for combining learning-aware anomaly detection with formal threat modeling in resilient CPS security operations.

Emphasis: AI-based threat modeling, adversarial intelligence, CPS resilience.

Index Terms: cyber–physical systems, SWaT, STRIDE, PASTA, LSTM, threat modeling, adversarial risk analysis, MITRE ATT&CK for ICS, ontology, resilience.

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1 Introduction

Cyber–physical systems (CPS) underpin industrial water treatment, power delivery, transportation, process control, and other safety-critical infrastructures. In such environments, cyber intrusions do not remain purely digital events; they can alter valves, pumps, chemical dosing units, sensor readings, and operator visibility, thereby producing immediate physical consequences [3, 4, 5, 12]. This direct linkage between cyber compromise and physical impact makes CPS security fundamentally different from conventional enterprise information security.

Threat modeling is widely used to analyze attack surfaces early in the design cycle. Frameworks such as STRIDE and PASTA help analysts organize threats, assign risks, and reason about attack paths [1, 2]. However, design-time models are typically static: they explain what could happen, but not how likely a threat becomes under changing runtime conditions. On the other hand, machine learning and deep learning techniques have shown strong promise in intrusion and anomaly detection for industrial control systems (ICS), especially when time-series telemetry must be interpreted sequentially [7, 8, 13]. Yet, purely data-driven detectors often act as black boxes and may not naturally support semantic threat interpretation, attack traceability, or operator-oriented mitigation planning.

This project proposes and implements an integrated framework that combines the strengths of both viewpoints. The core idea is simple but powerful: use structured threat modeling to define what matters, and use adversarial learning to continuously estimate when threat conditions are likely to be active. The SWaT testbed is used as the experimental platform because it is one of the most widely studied CPS security benchmarks and provides realistic water treatment telemetry under both normal and attacked operation [5, 6, 9].

The complete workflow was implemented as a sequence of six notebooks and corresponding exported artifacts. The flow begins with CPS decomposition and threat identification, proceeds through telemetry preprocessing and LSTM-based learning, and culminates in dynamic risk estimation, comparative analysis, and explainability through MITRE ATT&CK for ICS and ontology-based contextualization. In this sense, the project is framed not as a narrow anomaly detector, but as a complete cyber–physical security analytics pipeline that links system understanding, adversarial learning, risk reasoning, and contextual defense intelligence.

The major contributions of the project are summarized below.

1. A design-time CPS threat modeling pipeline for SWaT covering layered architecture, assets, STRIDE threats, attack trees, and baseline risk scoring.

2. A data preparation workflow for SWaT telemetry that converts raw process measurements into normalized temporal windows suitable for sequence learning.

3. A stacked LSTM-based adversarial behavior learner that produces high-quality attack probability estimates from multivariate CPS telemetry.

4. A hybrid STRIDE–LSTM risk engine that updates threat likelihoods over time and computes dynamic risk scores aligned with component-level cyber–physical threats.

5. A comparative analysis across standalone learning, LSTM+PASTA, and LSTM+STRIDE reasoning modes, with a static STRIDE-only baseline retained for reference.

6. An explainability layer using MITRE ATT&CK for ICS mapping, ontology construction, attack narratives, and operator-facing threat summaries.

The report follows the same sequence as the notebooks in order to preserve methodological traceability and to present the project as one continuous research pipeline from system understanding to security decision support.

2 Related Work

Cyber–physical systems security has evolved as a distinct research area because conventional information-security abstractions are insufficient for systems whose digital decisions directly govern physical processes. Foundational CPS research emphasizes that computation, networking, control, and safety constraints must be analyzed jointly rather than in isolation [3]. Later survey work on CPS security formalized this argument by showing that threats in CPS arise across heterogeneous layers, including sensing, communication, control logic, and human interaction [4]. In industrial settings, this viewpoint is reinforced by standards-oriented guidance such as NIST SP 800-82 Rev. 3, which highlights the tight coupling among reliability, availability, safety, and security in operational technology environments [10].

Threat modeling has long been used to embed security reasoning into system design. STRIDE provides a component-centric perspective that classifies threats into spoofing, tampering, repudiation, information disclosure, denial of service, and elevation of privilege [1]. PASTA extends this idea toward a risk-centric and attacker-aware process in which business impact, attack simulation, and threat decomposition are considered in a staged manner [2]. In industrial control contexts, these frameworks are attractive because they support structured attack reasoning, trust-boundary identification, and consequence-oriented analysis. However, their outputs are typically static and are not inherently coupled with online adversarial evidence.

Within industrial control systems, the SWaT testbed has become a benchmark platform for security experimentation and data-driven detection research. Mathur and Tippenhauer introduced SWaT as a six-stage water treatment testbed for ICS security research and training [5]. Goh et al. subsequently released the public dataset that captures both normal and attack behavior, making SWaT one of the most influential resources for reproducible CPS security experiments [6]. Broader surveys of testbeds and datasets consistently identify SWaT as a

major reference point for industrial anomaly detection benchmarking and resilience-oriented validation [9].

From the detection perspective, sequential learning and physics-aware monitoring have both shown strong potential. LSTM networks were introduced by Hochreiter and Schmidhuber to capture long-range temporal dependencies in sequence data [7]. In CPS and ICS settings, these recurrent models are attractive because attacks often manifest as evolving deviations rather than isolated point anomalies. Survey studies confirm that LSTM, GRU, autoencoder, and hybrid deep-learning architectures are among the most promising families for ICS attack detection [8]. Complementing purely data-driven methods, Giraldo et al. reviewed physics-based attack detection and highlighted the importance of incorporating process behavior into security monitoring [13]. Work by Adepu and Mathur on distributed attack detection further demonstrates the value of plant-aware intelligence in SWaT-like environments [14].

At the same time, explainability remains a major requirement in operational technology environments. Industrial operators do not only need a binary alert; they need to know which component is implicated, what attack technique is plausible, and how the alert aligns with known adversarial behaviors. MITRE ATT&CK for ICS provides a structured knowledge base of tactics and techniques specific to industrial environments [11]. Together with case analyses of incidents such as Stuxnet, this motivates the use of semantic threat knowledge to connect model outputs with recognizable adversarial procedures and actionable mitigation pathways [12].

Complementary official resources from NIST, CISA, and iTrust further reinforce the relevance of trustworthy CPS engineering, OT security practice, and realistic SWaT-based experimentation in this domain [15, 16, 17].

The present project positions itself at the intersection of these lines of work. Rather than choosing between static threat modeling and sequential learning, it unifies them in a single workflow. The result is a learning-aware threat modeling pipeline specifically tailored for resilient CPS analysis, with emphasis on dynamic risk estimation, comparative benchmarking, and operator-oriented explainability.

3 Methodology

This section describes the methodology exactly in the same sequence as the implemented notebooks. The goal is to make the end-to-end project flow explicit, starting with CPS system understanding and ending with contextualized threat intelligence.

3.1 Overall Workflow and Notebook Sequence

The project was organized as a six-stage workflow. Each notebook produces artifacts that are reused by the next stage, forming a consistent research pipeline rather than isolated experiments.

Table 1: Notebook-wise methodology flow used in the project

Stage	Notebook	Main output of the stage
N0	**CPS and Threat Modeling**	CPS decomposition, asset inventory, STRIDE threats, attack tree, baseline risk model
N1	**Data Preprocessing**	Cleaned SWaT telemetry, normalized features, sliding windows, metadata, train/test arrays
N2	**LSTM Adversarial Learning**	Trained LSTM model, probability outputs, threshold analysis, classifier evaluation
N3	**STRIDE–LSTM Integration**	Dynamic threat likelihoods, time-varying risks, system risk trends, threat rankings
N4	**Comparative Evaluation**	Standalone and integrated framework comparison, resilience analysis, lead-time analysis
N5	**MITRE and Ontology Context**	MITRE ATT&CK for ICS mapping, ontology graph, narratives, operator summary

Figure 1 illustrates the conceptual schema of the complete hybrid framework. The design-time knowledge branch provides baseline likelihoods and structural threat semantics, while the runtime learning branch provides attack probabilities from live telemetry. Their fusion yields dynamic risk scores that support prioritization, explainability, and resilience-oriented response.

The framework in Figure 1 can be interpreted as a tightly coupled four-layer pipeline:

- **System understanding layer:** SWaT is decomposed into assets, trust boundaries, process stages, and communication links so that cyber exposure is grounded in physical process knowledge.
- **Threat intelligence layer:** STRIDE analysis, attack trees, and attack graphs establish the adversarial hypotheses that define what should be monitored and prioritized.
- **Learning layer:** cleaned telemetry windows are fed into the stacked LSTM so that temporal evidence of attack behavior can be quantified as an adversarial probability signal.
- **Risk and explainability layer:** probability outputs are fused with design-time threat priors to generate dynamic risks, comparative resilience analysis, MITRE ATT&CK mappings, and ontology-backed operator narratives.

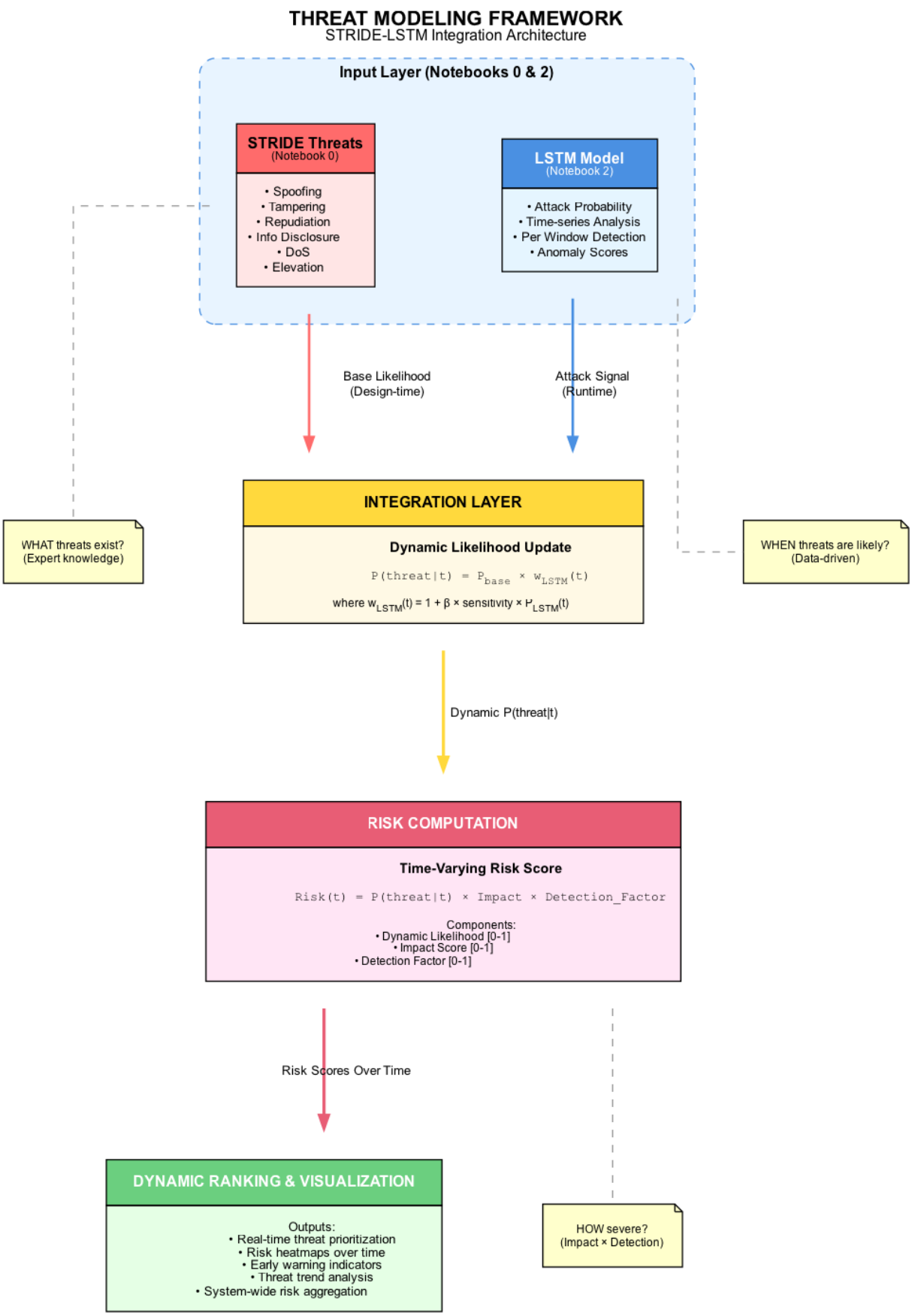

Figure 1: Integrated STRIDE–LSTM threat modeling framework used as the core project schema

3.2 CPS Architecture Modeling and Design-Time Threat Analysis

The first methodological stage modeled the SWaT testbed as a cyber–physical system with explicit physical, cyber, communication, and human layers. This decomposition was used to identify critical control paths, trust boundaries, and attack surfaces before any data-driven modeling was performed.

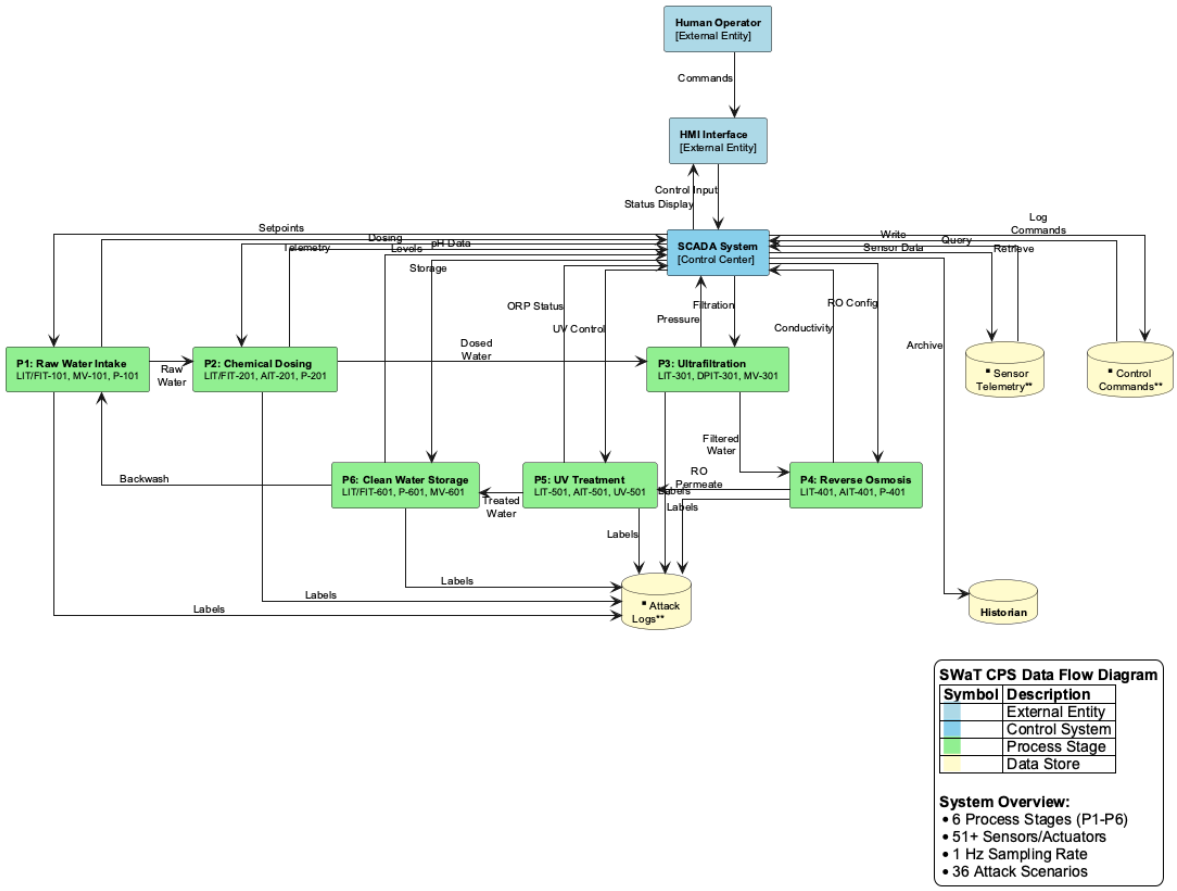

Figure 2: SWaT data flow diagram used to identify process stages, data stores, and trust boundaries

Table 2: Layered CPS view adopted for SWaT threat modeling

Layer	Main components	Primary security focus
Physical	Tanks, pumps, valves, sensors, actuators	Safety, process continuity, physical integrity
Cyber	PLCs, SCADA, HMI, embedded controllers	Command integrity, access control, logic protection
Communication	Modbus, EtherNet/IP, switches, control links	Authenticity, availability, traffic trustworthiness
Human	Operators, engineers, maintenance personnel	Authentication, authorization, accountability

The SWaT infrastructure was further annotated with component-specific attack surfaces to create a richer threat map for cyber–physical reasoning. The generated map emphasizes which

sensors, actuators, dosing pumps, and supervisory elements represent attractive adversarial targets.

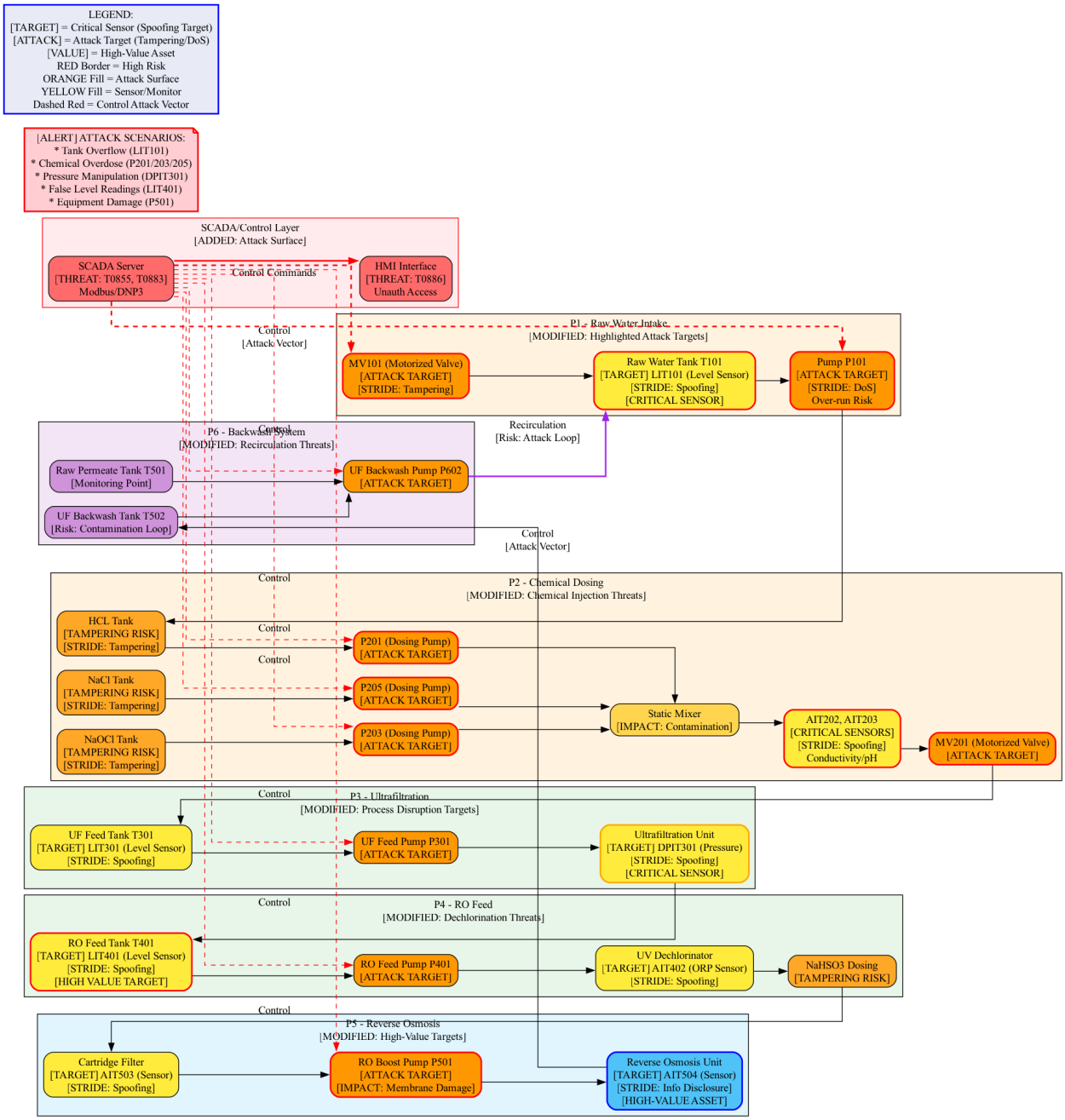

Figure 3: SWaT CPS infrastructure and threat map highlighting critical attack surfaces

STRIDE was then applied to key components including PLCs, SCADA, HMI, sensor networks, and the control network. From this catalog, eight representative threats were selected for baseline risk analysis. Each threat received a base likelihood, impact score, and detection difficulty factor. The baseline risk score used throughout the project was computed as

$$R_i^{(0)} = L_i^{(0)} \times I_i \times D_i \tag{1}$$

where $L_i^{(0)}$ is the design-time likelihood of threat i, I_i is its impact score, and D_i is a detection

difficulty multiplier.

Table 3: Top baseline STRIDE threats used for downstream dynamic risk analysis

ID	Threat description	Component	STRIDE	Risk score
T2	Spoofed sensor readings causing unsafe control	Sensor Network	Spoofing	1.500
T3	SCADA credential theft via phishing	SCADA	Spoofing	1.440
T5	Elevation of privilege on HMI workstation	HMI	Elevation	1.440
T1	Malicious PLC ladder logic injection	PLC	Tampering	0.900
T4	Denial of service against control network	Control Network	DoS	0.800
T6	Tampering with historian data	SCADA	Tampering	0.200
T7	Physical destruction of sensor network	Sensor Network	DoS	0.150
T8	Information disclosure of control algorithms	PLC	Info Disclosure	0.144

To complement the tabular threat model, the methodology also included structured attack reasoning through an attack tree and an attack graph. The attack tree focused on the concrete goal of causing a P1 tank overflow via logic manipulation, sensor spoofing, or protection bypass. The attack graph broadened this analysis to multi-stage cyber intrusion paths from initial access to physical impact.

3.3 SWaT Data Preprocessing and Feature Construction

The second notebook transformed raw SWaT telemetry into a sequence-learning dataset. The merged dataset was selected as the primary source, and its feature structure was analyzed to distinguish timestamp fields, attack labels, actuator states, and continuous sensor measurements. Labels were normalized into binary form, with 0 indicating normal operation and 1 indicating attack windows.

Preprocessing included missing-value handling, duplicate removal, outlier awareness, feature typing, normalization, and temporal window creation. Continuous features were scaled using MinMax normalization, while binary actuator channels were preserved in their operational form. This retained the semantics of on/off actuation while normalizing analog sensor ranges for learning stability.

The methodology also explored temporal feature engineering through first differences and rolling statistics, and stage-level feature aggregation through process-stage means, standard deviations, and maxima. These operations enriched the project's analytical understanding of SWaT behavior and informed interpretation, even though the final deployed LSTM operated on the core 51-feature telemetry representation.

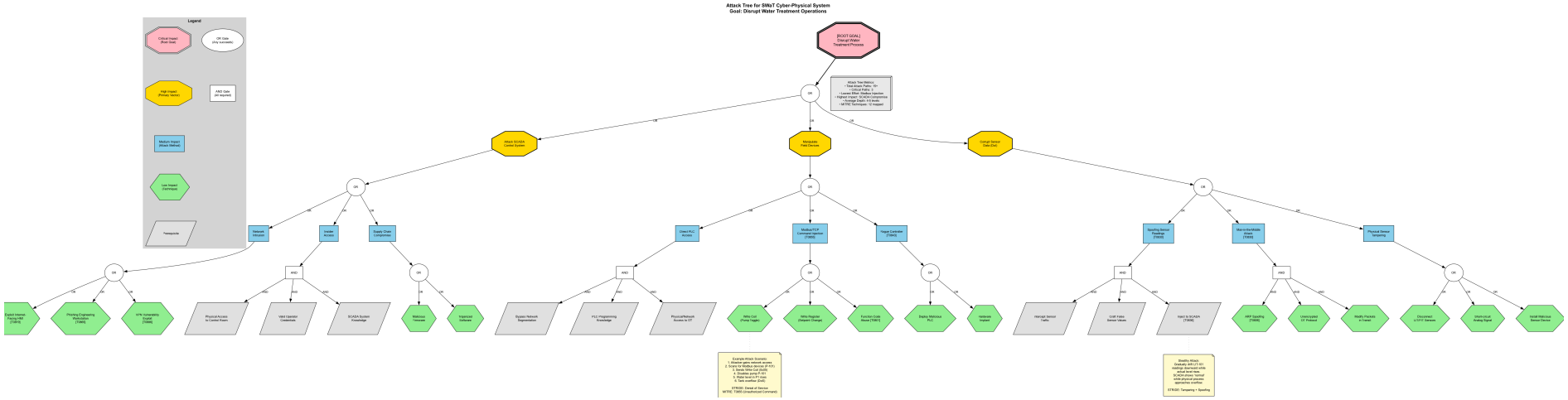

Figure 4: Attack tree used to model a representative unsafe process objective in SWaT

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Table 4: SWaT data preparation summary used for LSTM training

Category	Description	Value
Dataset scale	Raw merged samples read from the SWaT source file	1,441,719
Dataset scale	Cleaned samples retained for modeling after de-duplication	946,728
Dataset scale	Duplicate rows removed during preparation	494,991
Feature space	Final telemetry channels used for the deployed LSTM	51
Feature space	Binary actuator/state features	26
Feature space	Continuous sensor/process features	25
Feature engineering	Temporal analytical features generated for interpretation	25
Feature engineering	Stage-wise aggregate features generated for interpretation	12
Windowing setup	Sliding window size adopted for sequence learning	30 time steps
Windowing setup	Window stride used across the time series	1
Windowing setup	Total windows generated from the normalized dataset	946,699
Experimental split	Training windows used for model optimization	662,689
Experimental split	Testing windows used for final evaluation	284,010
Label distribution	Post-cleaning normal samples	892,107
Label distribution	Post-cleaning attack samples	54,621

Sliding windows were created from the normalized data using

$$\mathbf{X}_k = [\mathbf{x}_k, \mathbf{x}_{k+1}, \ldots, \mathbf{x}_{k+w-1}], \qquad y_k = \ell_{k+w-1} \tag{2}$$

where $w = 30$ and ℓ_{k+w-1} is the attack label associated with the final time step of the window. This formulation allowed the model to learn short-horizon temporal dependencies that precede attack states while preserving fixed-size sequence inputs.

3.4 LSTM-Based Adversarial Behavior Learning

The third notebook implemented a stacked LSTM network for binary attack detection over SWaT telemetry. LSTM was selected because CPS attacks often manifest as temporal deviations in sequential sensor behavior rather than isolated point anomalies [7, 8]. The adopted architecture used two recurrent layers followed by a dense fusion layer and a sigmoid output unit.

Table 5: LSTM architecture and optimization configuration

Configuration item	Value
Input shape	30×51
Recurrent architecture	Stacked LSTM
First LSTM layer	64 units, return sequences enabled
Second LSTM layer	32 units
Dense layer	16 ReLU units
Output layer	1 sigmoid unit
Dropout	0.2 after each LSTM block
Optimizer	Adam (10^{-3} learning rate)
Loss function	Binary cross-entropy
Batch size	64
Epoch budget	50 epochs
Validation split	20% of training set
Callbacks	Early stopping, model checkpointing, learning-rate reduction
Total parameters	42,657

Class imbalance was addressed through balanced class weighting computed from the training labels. This ensured that attack windows, which are much less frequent than normal operation windows, still exerted strong influence during optimization. The detector outputs an attack probability

$$\hat{p}_t = \sigma\left(\mathbf{W}\mathbf{h}_t + b\right) \tag{3}$$

for each input window, where $\mathbf{h}_t$ is the learned hidden representation and $\sigma(\cdot)$ is the sigmoid activation.

Threshold analysis was then applied over the predicted probabilities to compare the default 0.5 decision threshold against the best F1-operating point. The final exported artifacts included the trained model, probability arrays, confusion matrix, ROC and precision–recall curves, training history, and JSON summaries for use in later integration stages.

3.5 Learning-Aware STRIDE Risk Integration

The fourth notebook fused the LSTM attack probability stream with the design-time STRIDE threat model. The fusion logic assigns each STRIDE category a sensitivity factor that controls how strongly runtime anomaly evidence should modulate the baseline threat likelihood.

The runtime probability stream was first smoothed using a Gaussian filter with $\sigma = 5$ to obtain a stable adversarial activity profile. For each threat i, the baseline likelihood $L_i^{(0)}$ was updated at time t according to

$$L_i(t) = \min\left(1,\ L_i^{(0)}\left[1 + \beta s_i \hat{p}_t\right]\right) \tag{4}$$

Table 6: STRIDE sensitivity weights used for dynamic likelihood updating

STRIDE category	Sensitivity weight
Tampering	2.0
Spoofing	1.5
DoS	1.2
Elevation	1.0
Repudiation	0.5
Info Disclosure	0.3

where s_i is the sensitivity associated with the threat category and $\beta = 1.0$ is a global influence factor.

The time-varying threat risk was then computed as

$$R_i(t) = L_i(t) \times I_i \times D_i \tag{5}$$

and the overall system risk was aggregated as

$$R_{\text{system}}(t) = \sum_{i=1}^{N} R_i(t) \tag{6}$$

over the eight modeled threats. This produced dynamic threat rankings, risk heatmaps, aggregate system risk traces, and lead-time statistics that connect sequential anomaly evidence with component-level cyber–physical threat semantics.

3.6 Comparative Analysis Design

The fifth methodological stage evaluated the framework from a comparative perspective. Three main operational views were considered:

1. **Standalone LSTM**: a purely data-driven attack detector producing attack probabilities and binary labels.
2. **LSTM+PASTA**: the same detector enriched with attacker-centric and quantified risk context.
3. **LSTM+STRIDE**: the same detector enriched with component-centric threat types and dynamic risk prioritization.

In addition, a static STRIDE-only baseline was retained as a conceptual benchmarking reference for contextual performance plots. Comparative evaluation considered detection quality,

false-positive rate, early warning lead time, threat categorization, explainability, and a composite resilience score.

Table 7: Evaluation dimensions used in the comparative study

Dimension	Interpretation in this project
Detection accuracy	Correct overall classification of attack and normal windows
Precision and recall	Ability to balance false alarms and missed attacks
F1-score and ROC–AUC	Overall detector quality under thresholded and ranking views
Lead time	Number of time steps by which attacks are signaled before onset
Threat categorization	Whether the framework attaches explicit threat semantics to alerts
Risk prioritization	Whether the framework offers static, quantified, or dynamic threat ranking
Explainability	Degree of semantic context available to support operator interpretation
Resilience score	Composite score combining detection, false alarms, and contextual strength

3.7 MITRE ATT&CK for ICS Mapping and Ontology Context

The final notebook added semantic depth to the runtime outputs. Each baseline threat was mapped to MITRE ATT&CK for ICS tactics and techniques, thereby translating model alerts into known adversarial behaviors. The project covers 16 distinct MITRE techniques across 8 modeled threats.

A lightweight directed ontology was then constructed with four concept classes: system components, STRIDE categories, MITRE techniques, and operational impacts. Edges encode how components are threatened, how STRIDE classes cause impact, and how adversarial techniques are associated with threat categories. Centrality analysis, attack path search, and operator summary tables were generated from this graph to support explainable and actionable security reporting.

4 Experiments and Results

This section presents the experimental outcomes produced by the implemented notebooks and exported artifacts. All figures and numerical summaries are drawn from the project outputs generated in the workflow.

4.1 Threat Modeling Outputs

The design-stage analysis yielded a structured threat model that connects SWaT components, attack surfaces, and process consequences. The attack graph in Figure 5 demonstrates how cyber intrusion can progress from external access through SCADA compromise, PLC manipulation, sensor spoofing, and finally physical process impact. This is particularly important in CPS security because it shows that detection should not be limited to isolated alerts; it must support multi-stage adversarial reasoning.

The baseline STRIDE risk table identified spoofed sensor readings (T2), SCADA credential theft (T3), and HMI privilege escalation (T5) as the most critical threats. This ranking established the anchor for all subsequent dynamic risk updates and comparative resilience analysis.

4.2 LSTM Training Behavior and Detection Performance

Training converged smoothly under the selected configuration. The exported training history shows stable loss reduction and strong validation behavior across the 50-epoch schedule.

The final LSTM model delivered strong classification performance on the test set. The confusion matrix and ranking curves confirm that the detector separates normal and attack windows with high confidence across threshold-based and threshold-independent views.

Table 8: Standalone LSTM attack detection performance

Metric	Value
Accuracy	0.9957
Precision	0.9334
Recall	0.9965
F1-score	0.9639
ROC–AUC	0.9999
PR–AUC	0.9985
Matthews correlation coefficient	0.9622
Cohen's kappa	0.9616
True negatives	266,459
False positives	1,165
False negatives	57
True positives	16,329
Default decision threshold	0.50
Optimal F1 threshold	0.88
Optimal F1-score	0.9806

The near-perfect ROC–AUC and PR–AUC values indicate that the model is highly effective in ranking attack windows above normal behavior. In practical terms, the model provides a

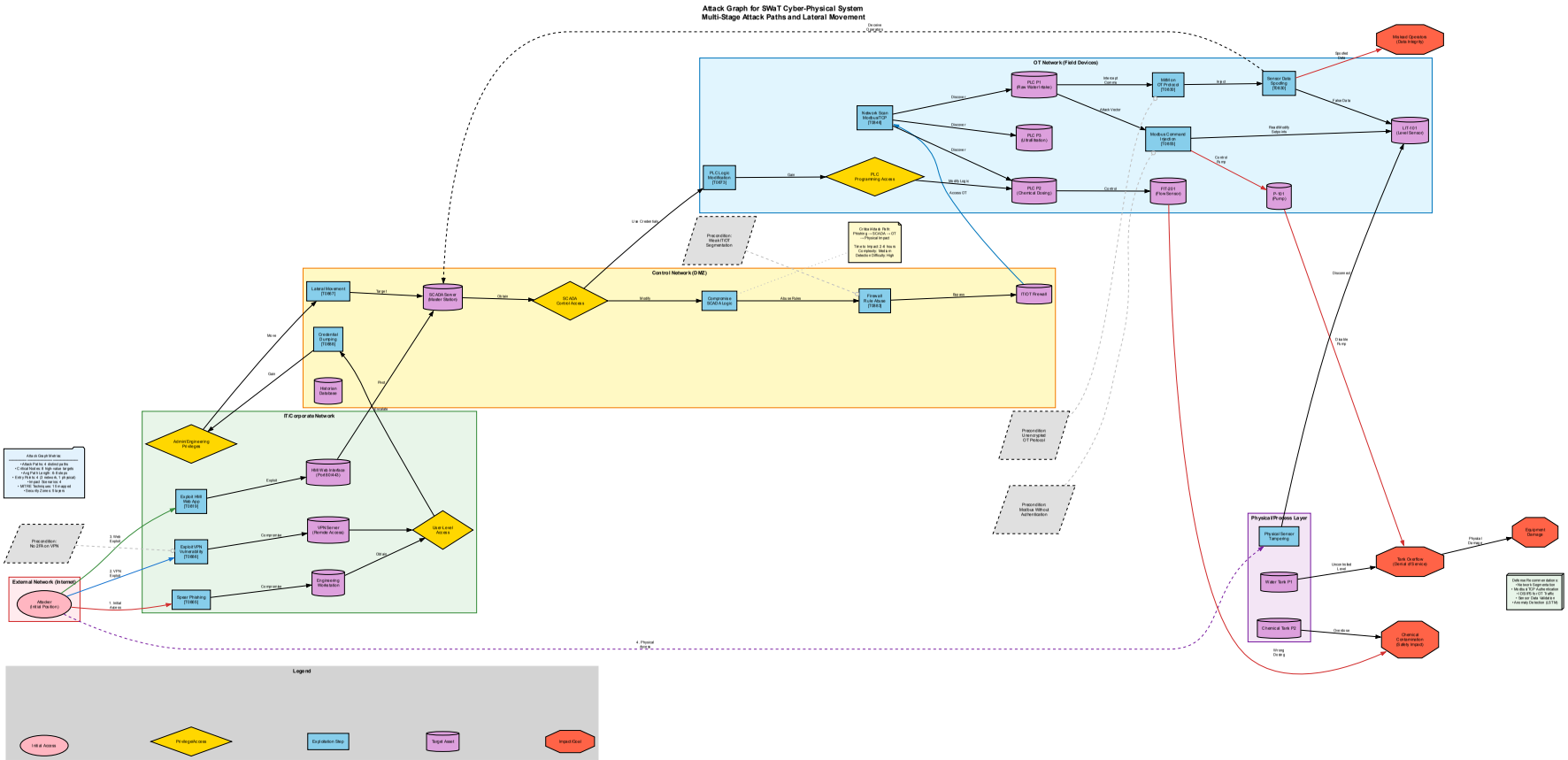

Figure 5: Attack graph illustrating multi-stage adversarial progression across the SWaT CPS

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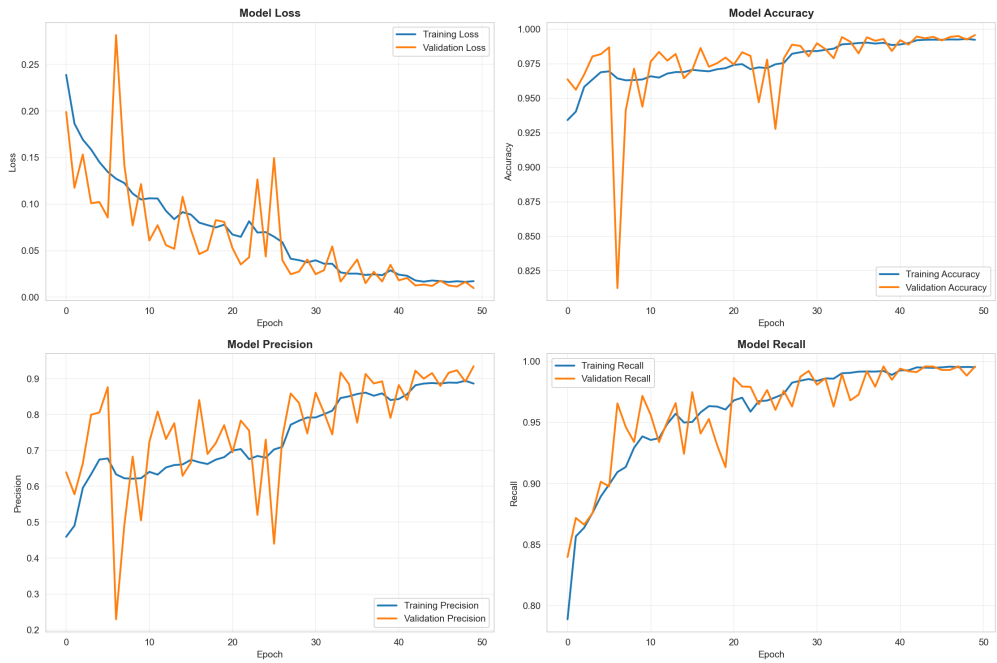
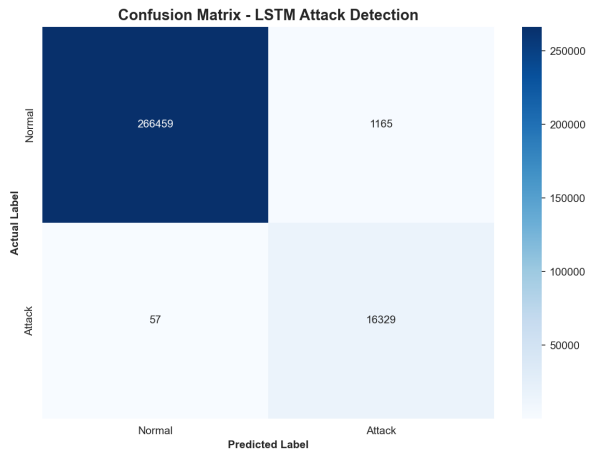
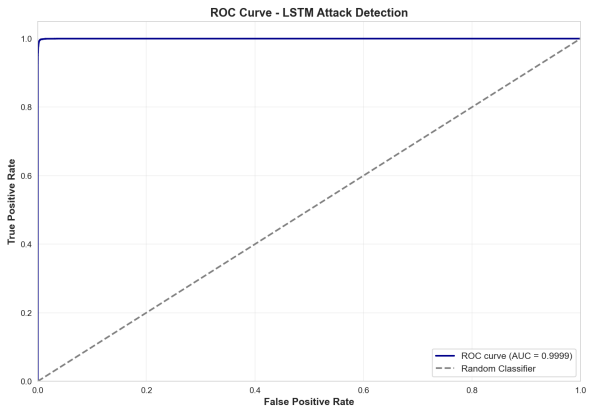


Figure 6: Training history of the stacked LSTM detector on SWaT telemetry windows



(a) Confusion matrix



(b) ROC curve

Figure 7: Core classification results of the LSTM detector

reliable adversarial activity signal that can be used not only for classification, but also for downstream dynamic risk modulation.

4.3 Dynamic Risk Assessment Results

The hybrid STRIDE–LSTM fusion stage converts model confidence into component-level threat intensity. Figure 8 shows how the aggregate system risk rises above the static baseline during attack conditions, while the heatmap in Figure 9 shows time-varying behavior across all modeled threats.

The highest-priority threats remained T2, T3, T5, T1, and T4 throughout the study, but

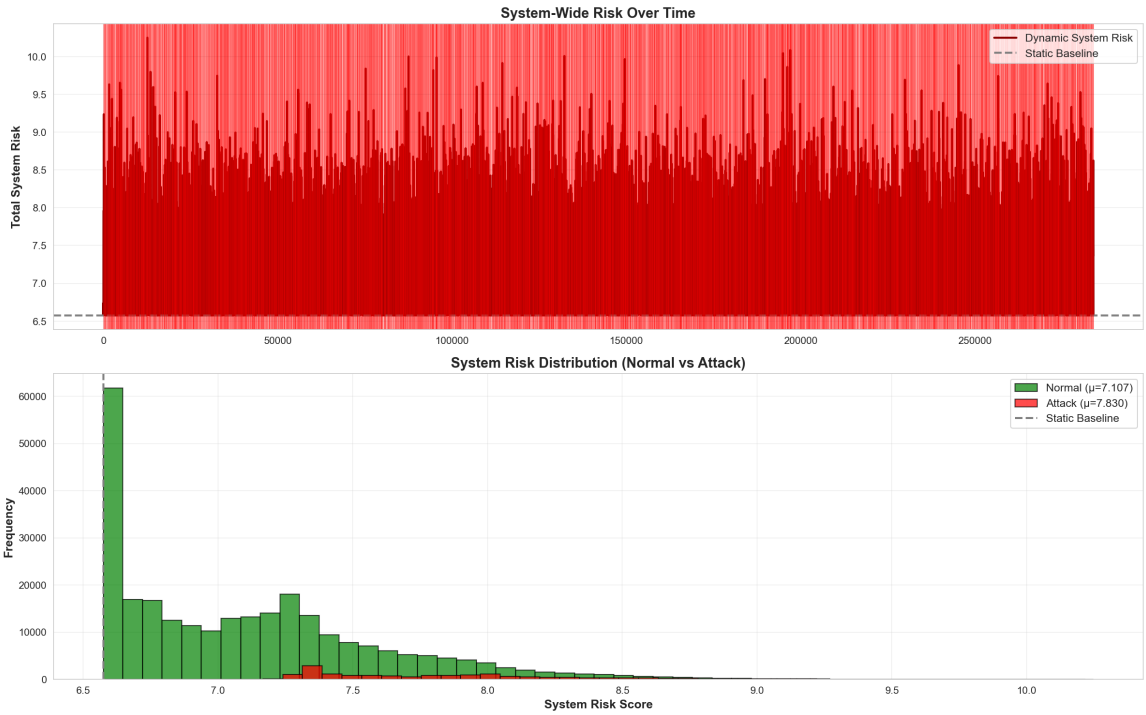


Figure 8: System-wide static versus dynamic risk behavior under normal and attack periods

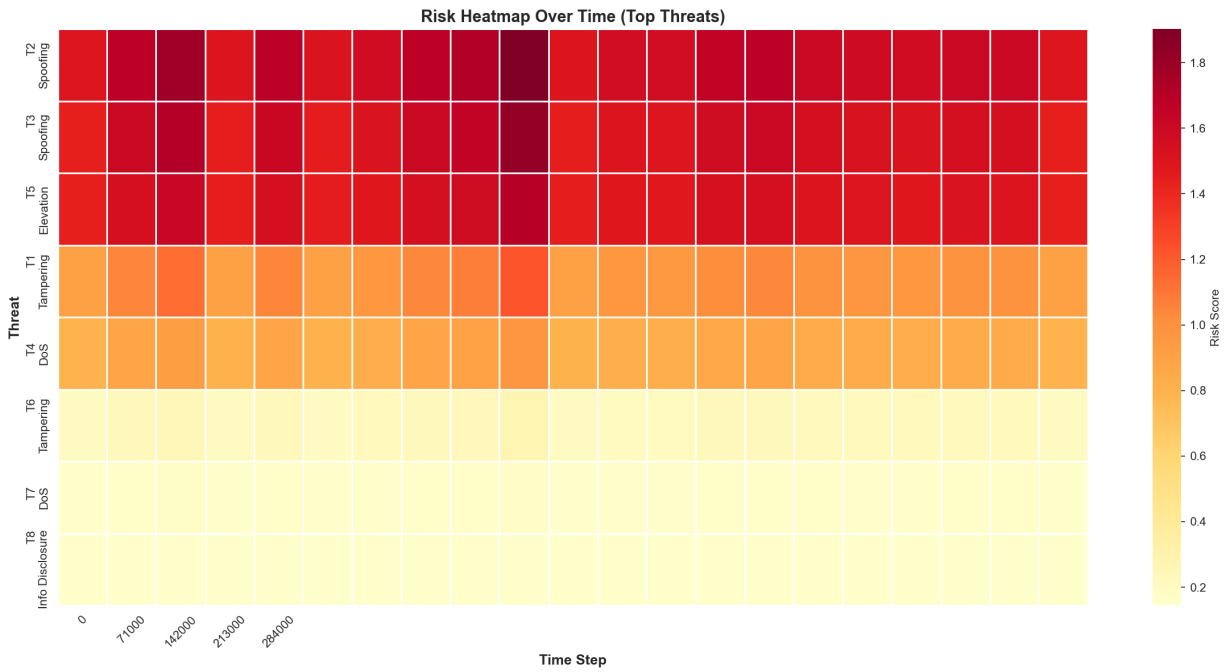


Figure 9: Threat-wise dynamic risk heatmap across the analyzed timeline

Table 9: System-wide risk statistics produced by the integrated STRIDE–LSTM model

Risk quantity	Value
Static baseline total risk	6.5740
Dynamic risk mean (overall)	7.1490
Dynamic risk median (overall)	7.0798
Dynamic risk standard deviation	0.5173
Dynamic risk minimum	6.5740
Dynamic risk maximum	10.2482
Dynamic risk mean during normal periods	7.1073
Dynamic risk mean during attack periods	7.8297
Risk increase during attack periods	10.16%

their risk magnitudes increased adaptively as the LSTM signal strengthened. This confirms that the integrated model acts as a real-time prioritization engine built on top of design-time cyber–physical threat knowledge.

4.4 Comparative Analysis

Comparative analysis was treated as a dedicated result block because the project aims not only to detect attacks but also to show why integration produces more operationally useful security reasoning.

Table 10 presents the requested comparison among standalone LSTM, LSTM+PASTA, and LSTM+STRIDE. The standalone detector provides strong attack classification. LSTM+PASTA adds attacker-centric quantified risk context. LSTM+STRIDE retains the strong detector while also introducing explicit threat categories, component-oriented interpretation, and dynamic risk prioritization. For visual reference, the original comparison stage also retained a static STRIDE-only baseline to contrast pure design-time reasoning against integrated learning-aware methods.

Table 10: Clear comparative analysis among standalone LSTM, LSTM+PASTA, and LSTM+STRIDE

Approach	Acc.	Prec.	Rec.	F1	ROC–AUC	Distinct advantage
Standalone LSTM	0.995	0.930	0.994	0.961	0.999	Strong time-series attack detection from telemetry windows
LSTM+PASTA	0.995	0.932	0.995	0.963	0.999	Adds attacker-centric and quantified risk interpretation
LSTM+STRIDE	0.996	0.936	0.997	0.966	1.000	Adds explicit threat categories, dynamic prioritization, and high explainability

Table 11: Framework-level comparison with the static STRIDE-only reference baseline

Evaluation dimension	STRIDE-only	PASTA+LSTM	STRIDE+LSTM
Detection accuracy	0.942	0.996	0.996
Detection precision	0.000	0.932	0.936
Detection recall	0.000	0.995	0.997
F1-score	0.000	0.963	0.966
ROC–AUC	0.500	0.999	1.000
False positive rate	0.000	0.004	0.003
Early detection lead time	0.0 steps	15.8 steps	16.9 steps
Threat categorization	6 categories	6 quantified threats	8 threats
Risk prioritization	Static	Risk-centric	Dynamic
Explainability	High	High	High
Resilience score	0.000	0.794	0.992

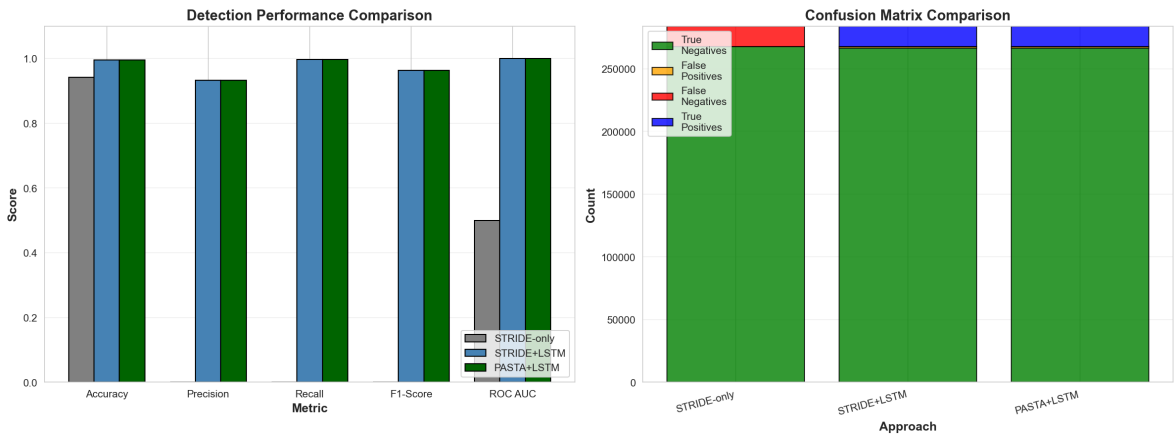
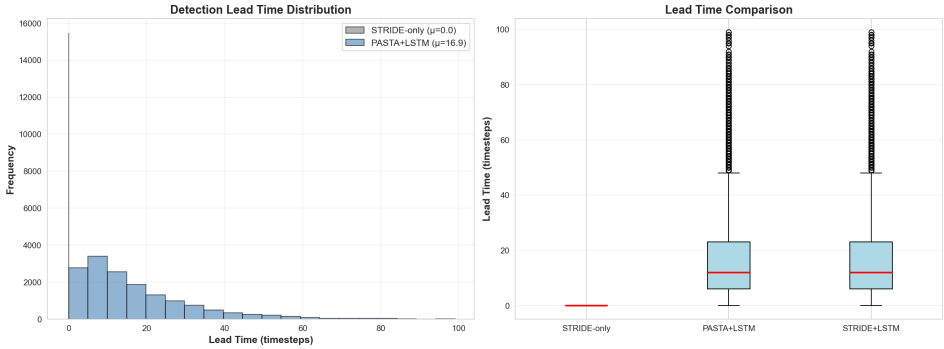


Figure 10: Comparative performance view across the three framework configurations and the static reference baseline

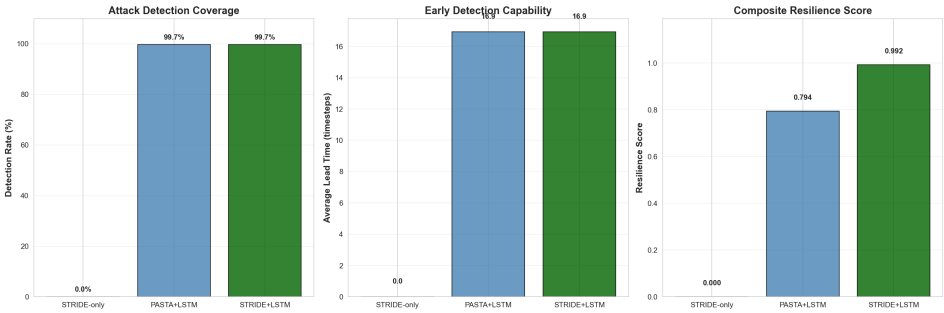
The comparative section makes two observations especially clear. First, the hybrid models preserve the high-quality sequence learning capability of the LSTM detector. Second, the addition of threat modeling creates a meaningful jump in operational usefulness by turning prediction scores into structured, prioritized, and explainable security insight. In the final resilience scoring, LSTM+STRIDE attains 0.992, exceeding the 0.794 achieved by LSTM+PASTA and vastly outperforming the static STRIDE-only baseline.

4.5 MITRE ATT&CK and Explainability Results

The explainability layer transformed high-confidence alerts into operator-oriented security intelligence. The project generated a threat-to-technique mapping table, MITRE tactic distribution plots, an ontology graph, centrality analysis, and example attack narratives.



(a) Lead-time comparison

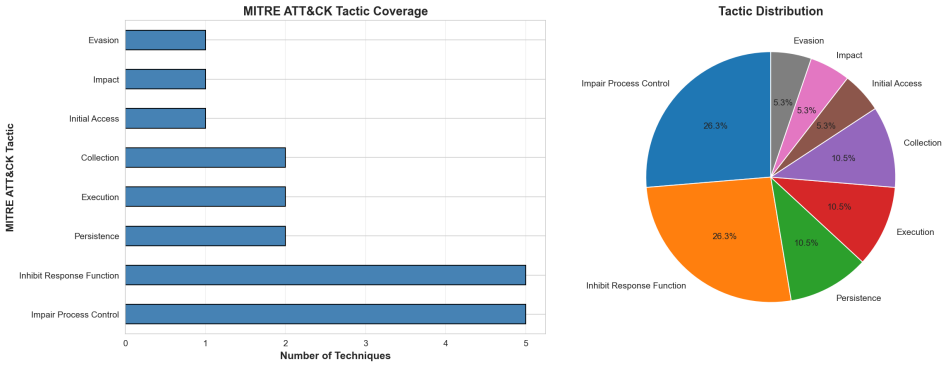


(b) Resilience comparison

Figure 11: Temporal advantage and resilience gain of the integrated framework

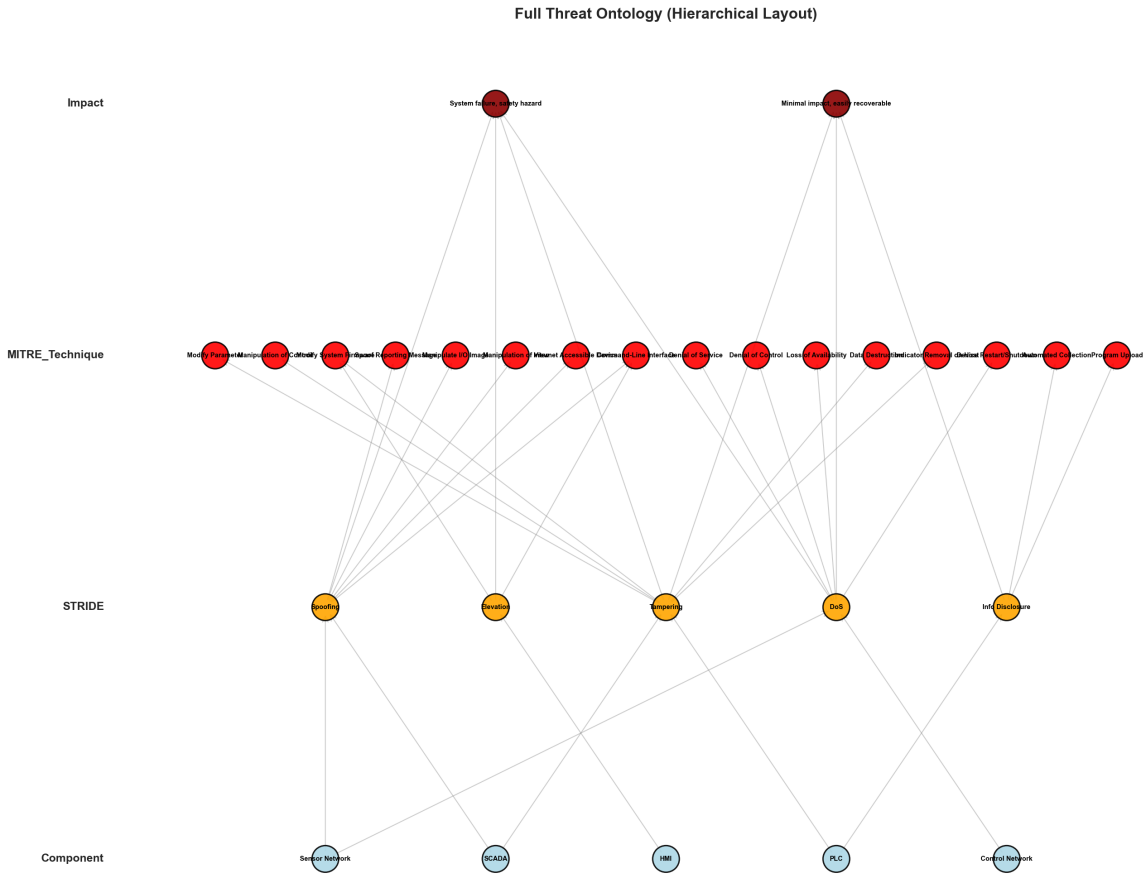
Table 12: Explainability and contextualization summary

Category	Indicator	Result
ATT&CK mapping	STRIDE threats linked to MITRE ATT&CK for ICS techniques	8
ATT&CK mapping	Unique ATT&CK techniques covered by the mapping	16
Knowledge graph	Ontology nodes representing assets, threats, techniques, and impacts	28
Knowledge graph	Directed edges encoding semantic security relationships	33
Semantics	Node types included in the ontology	Components, STRIDE classes, ATT&CK techniques, impacts
Semantics	Most central node in the graph-based analysis	Spoofing
Narrative output	Representative high-priority narrative instance	T2 (sensor spoofing)
Narrative output	Example analyst confidence level	High (LSTM probability 1.0)



(a) MITRE tactic distribution

Figure 12: MITRE ATT&CK for ICS contextualization and ontology construction



(b) Threat ontology overview

Figure 12: MITRE ATT&CK for ICS contextualization and ontology construction (continued)

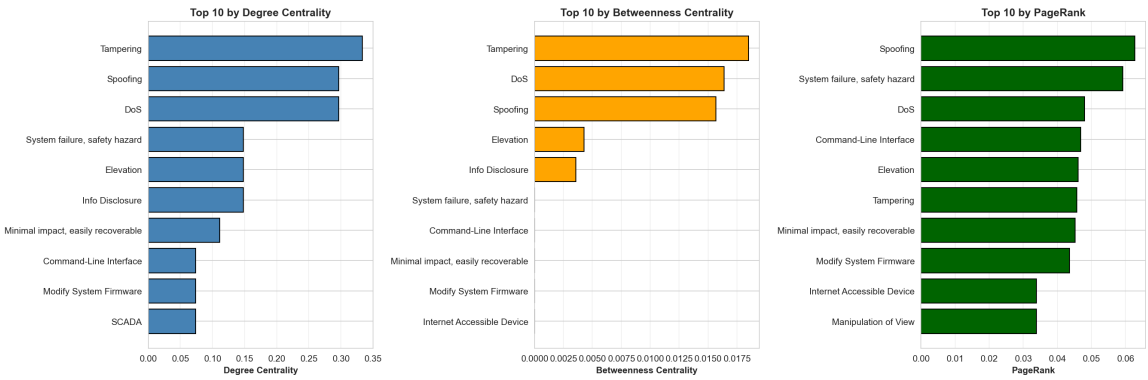


Figure 13: Centrality analysis of the lightweight threat ontology

These outputs are valuable because they bridge the gap between a high numerical anomaly score and a human-usable operational interpretation. In particular, the ontology views make it possible to move from a detected anomaly to a semantically organized explanation that ties affected components, threat categories, ATT&CK techniques, and operational consequences into one coherent analyst-facing representation.

5 Discussion

The experimental evidence supports several research-level observations about the value of learning-aware threat modeling for resilient CPS defense.

5.1 Key Technical Takeaways

- **The methodology is strongly layered and logically traceable.** Each notebook contributes a concrete analytical function, beginning with system understanding and threat enumeration, then progressing through temporal learning, dynamic risk fusion, comparative benchmarking, and semantic contextualization.
- **The learning component contributes high-confidence temporal discrimination.** The LSTM results show that multivariate telemetry windows can separate normal and attack behavior with very strong classification and ranking performance, making the probability stream suitable for downstream risk-aware reasoning.
- **The threat-modeling component preserves cyber–physical meaning.** STRIDE categories, component mappings, and attack-structure artifacts ensure that alerts are not treated as isolated scores but as threat instances with explicit operational semantics.
- **The fusion layer creates actionable runtime intelligence.** By converting model confidence into threat-conditioned risk, the framework moves from detection toward prioritization, which is one of the most important requirements in operational CPS defense.

5.2 Comparative Interpretation

- **Standalone LSTM is strong at detection,** but its primary output remains a confidence score and classification decision.
- **LSTM+PASTA improves attacker-oriented risk framing,** which is valuable for scenario-driven reasoning and quantified adversarial analysis.
- **LSTM+STRIDE provides the most balanced operational profile,** because it preserves high detection quality while adding explicit threat taxonomy, component-centric interpretation, and dynamic prioritization. This makes it the strongest configuration in the comparative evaluation.
- **The comparative block strengthens the contribution of the report,** because it shows that the work is not merely about obtaining a good classifier, but about demonstrating how different security reasoning frameworks shape practical cyber–physical defense value.

5.3 Practical and Research Significance

- **For practitioners,** the framework offers a pathway from raw telemetry to interpretable, triage-ready security insight.
- **For research evaluation,** the project demonstrates breadth and depth by integrating CPS modeling, threat analysis, adversarial learning, comparative security analytics, and explainability in one coherent pipeline.
- **For resilient operations,** the MITRE ATT&CK and ontology layers are especially important because they convert numerical anomaly evidence into analyst-facing narratives, semantic relationships, and mitigation-oriented context.

6 Conclusion

This report presented a complete research-oriented realization of *AI-driven threat modeling for resilient cyber–physical systems* over the SWaT water treatment testbed. The work contributes a coherent end-to-end pipeline in which design-time CPS understanding, STRIDE-based adversarial reasoning, telemetry-centric sequence learning, and semantic explainability are not treated as isolated modules, but as mutually reinforcing parts of one integrated security analytics framework.

The experimental findings demonstrate that this integration is technically effective and analytically meaningful. The stacked LSTM detector achieved 99.57% accuracy with a ROC–AUC of

0.9999, confirming strong temporal discrimination over multivariate SWaT telemetry. When this learning signal was fused with the STRIDE threat model, the resulting framework produced dynamic risk evolution, clearer threat prioritization, stronger resilience-oriented interpretation, and richer operator context than standalone detection alone. Comparative analysis further showed that the LSTM+STRIDE configuration delivers the most complete operational profile among the studied alternatives by combining high detection quality with explicit threat semantics and actionable interpretability.

Taken together, the project establishes a strong methodological blueprint for resilient CPS defense in which *what can go wrong*, *when it appears to be happening*, and *how it should be interpreted* are answered within a unified architecture. As a research contribution, it demonstrates how adversarial intelligence can be embedded into formal threat modeling to produce security outputs that are not only accurate, but also prioritized, explainable, and practically aligned with cyber–physical incident response.

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